

Department of Mathematical Sciences
Faculty of Science
Sa'adu Zungur University, Gadau

200 Level First Semester (2025/2026) Examination

Course Code: IFT 211 Course Title: Digital Logic Design Time: 02:00hrs

Instruction: Attempt Any Three Questions

Question 1

Simplify the following Boolean functions using K-Map:

(i) $F(A,B,C,D) = \Sigma(0,1,2,8,10,11,13)$ **[5 marks]**

(ii) $F(A,B,C,D) = \Sigma(0,1,2,3,5,8,10,14,15)$ **[5 marks]**

(iii) $F(W,X,Y,Z) = \Sigma(1,2,4,8,11,12,13)$ **[5 marks]**

(iv) $F(A,B,C) = \Sigma(0,1,5,7)$ **[5 marks]**

(Total 20 marks)

Question 2

(a) Fill up the following table. Show your workings clearly.

Binary	Octal	Hexadecimal
		FACE
0.17	143	
1001001		
	77.35	

(b) Design the Logic Circuit Diagram for a 4X1 Multiplexer. **(12 marks)**

(Total 20 marks)

Question 3

(a) Design a 3-input AND gate using the appropriate Multiplexer. **(10 marks)**

(b) Draw the logic circuit diagram that implements the following Boolean functions: **(10 marks)**

(i) $F(A,B,C,D) = ABCD + AB'C'D + ABC'D$

(ii) $F(x,y,z) = (x+y+z)(x'+y'+z)(x+y+z')$

(Total 20 marks)

Question 4

(a) Design a BCD-Decimal decoder using the unused combination of the BCD codes as don't care conditions. **(12 marks)**

(b) Convert the Boolean function $F(A,B,C) = AC' + BC + BC'$ into the standard canonical S.O.P and P.O.S form. **(8 marks)**

(Total 20 marks)

Question 5

(a) Implement the following Boolean Functions with a Multiplexer: **(15 marks)**

(i) $F(A,B,C,D) = \Sigma m(2,3,4,5,8,9,12)$

(ii) $F(A,B,C,D) = \Sigma m(0,2,5,7,11,14)$

(iii) $F(A,B,C) = \Sigma m(1,3,5,6)$

(b) State any five laws of Boolean Algebra. **(5 marks)**

(Total 20 marks)